



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 01

Objective & Lecture Mapping: Recall that an intelligent agent acts within a continuous loop: Sensors generate percept sequences, and Actuators execute actions to maximize a Performance measure. Use this sheet to execute practical modifications and incrementally evaluate your design against Lecture 01 and 02 theory (from basic recall to analytical classification).

Part 1: Code Analysis & PEAS Evaluation

Task 0 : Setting up and getting the starter Code

1. Go to the [Git Hub Repo](#). Create a Fork of this Git Repo to your Personal Github Profile. Open it using your favourite IDE and follow the below instruction to complete the practical. The base code is in the "master" brach of the repo.

Task 1.1: The Environment State (`__init__`)

1. (Remember) List the four components of the PEAS framework discussed in Lecture 01.

E - Environment
A - Actuators
S - Sensor



2. (Understand) Look at the `VisualGridHuntGame.__init__` method. Which specific variables in this code represent the physical "Environment (E)" state?

```
self.walls  
self.opponents  
self.agent_pos
```

3. (Analyze) Based on the variables initialized (specifically `self.opponents`), classify this baseline environment as "Single-Agent" or "Multi-Agent". Briefly justify your choice.

Multi Agent. There's main agent and multiple opponents in itself. These enemies move in grid individually and they can collide with the main agent.

Task 1.2: The Perception Subsystem (`get_percept`)

4. (Remember) Define what a "percept sequence" is according to Lecture 01.

A percept sequence is the complete history of all the percepts the agent has ever received. The agent uses this accumulated information to make decision on its next action.

5. (Understand) Which component of the PEAS framework does the `get_percept()` method represent in our code?

"S" - Sensors. It collects information from the environment.

6. (Evaluate) Based on the exact dictionary returned by `get_percept()` , is this environment "Fully Observable" or "Partially Observable"? Explain why based on what the agent can and cannot see.

It's fully observable. The agent can see its own position. where all the opponents are, whether it's on food, if there's collision, its score, and how much food is left.

Task 1.3: Action Execution (`execute_action`)

7. (Understand) When `execute_action()` deducts points for hitting a wall, which PEAS component is being actively updated?

Performance Measure. The score changes when the agent hits a wall, so the performance measure is what's being updated.

8. (Evaluate) Why is it structurally important that this metric evaluates changes in the external environment state (hitting a wall) rather than the agent's internal processing effort?

It's important, because a performance measure should penalize the agent for crashing into a wall. Penalty reflects how agent is navigating.(whether its good or bad).

Part 2: Guided Modification & Deep Theory Mapping

Follow the implementation instructions to inject a new hazard, then answer the questions to map your code changes back to the theoretical nature of the environment.

Step 2.1: Extending the Environment Initialization (`__init__`)

1. **Implementation Instructions:** Declare a new trap collection attribute (e.g., `self.toxic_traps = set()`) inside `__init__`. Populate it with coordinate tuples safely avoiding position `(0, 0)`, walls, and food.

9. (Understand) If you successfully add `self.toxic_traps` to the environment but intentionally **hide** this data from the agent's sensors, how does this specifically alter the environment's "Observability" classification?

The problem becomes partially observed because the agent cannot see where the toxic traps are. Agent can't know what it doesn't know.

Step 2.2: Updating the Perception Subsystem (`get_percept`)

1. **Implementation Instructions:** Locate `get_percept(self)` and add a new boolean sensor key (e.g., `'smells_toxin': tuple(self.agent_pos) in self.toxic_traps`) to the dictionary returned to the agent loop.

10. (Analyze) By adding `'smells_toxin'`, you expand the agent's percept sequence. Explain how this specific sensor helps the agent act with "Rationality" (maximizing expected utility).

Adding the `smells_toxin` sensor would give extra information to the agent. Using this information agent can avoid getting close to the traps.

Step 2.3: Modifying Action Execution & Visual Rendering (`execute_action` & `draw_grid`)

1. **Implementation Instructions:** In `execute_action`, check if the updated `self.agent_pos` intersects with `self.toxic_traps` to decrement `self.score` by 15 points. In `draw_grid`, iterate through traps to render custom purple shapes on the canvas.

11. (Remember) You programmed a severe score penalty for stepping on a trap to avoid metric exploitation. What was the classic "Vacuum World Exploit" example discussed in Lecture 01 that illustrated the danger of faulty metrics?

The example showed that if the performance measure isn't designed properly, the vacuum agent might find a loophole. Instead of just cleaning efficiently, it could keep cleaning the same spot just increase its score. It showcases why the

Finalize and Lock Lab 01 Analytical Evaluation

Lab 01 code analysis and theoretical evaluation complete and verified!



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